Facial Emotion Recognition Using PCA and Random Forest

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CSE 2600: Introduction to Data Science and Engineering

1. Introduction

Facial emotion recognition is a field of computer vision that involves interpreting the emotional state of a person based on their facial features. This problem sits at the intersection of machine learning, psychology, and human computer interaction, and has significant real-world consequences across a variety of domains. From accessibility tools that help people with social communication challenges to mental health monitoring systems capable of detecting signs of distress, the ability to classify human emotion from still images holds meaningful practical promise.

This project investigates whether classical machine learning methods, specifically Principal Component Analysis (PCA) and Random Forest classification, can produce meaningful results on this task. Rather than using neural networks that require a large number of computational resources, we wanted to understand how well traditional approaches can accomplish a similar task. We were particularly interested in examining whether dimensionality reduction through PCA would improve classification accuracy by removing noise and redundancy from raw pixel data, or whether it would ignore subtle features that are essential for distinguishing between emotional expressions. The findings from this analysis contribute to a broader understanding of when and why dimensionality reduction is beneficial in image classification models.

2. Dataset Description

2.1 Overview

The dataset used in this project is the FER-2013 (Facial Expression Recognition 2013) dataset, which is widely used for facial expression classification projects. It contains approximately 35,000 training images and 4,000 test images, which all stored as 48x48 pixel grayscale faces labeled across seven emotion categories: angry, disgust, fear, happy, neutral, sad, and surprise. The dataset was originally introduced as part of a Kaggle competition and has since been used extensively in published research. It contains pre-partitioned training and testing splits, eliminating excessive steps in setting up. Its relevance to our project goal and objective made it a good option to use.

2.2 Preprocessing

Several preprocessing steps were applied before modeling. First, each image was loaded and converted to grayscale. Then, pixel values from the images were normalized from a 0 to 255 range down to a 0 to 1 scale to keep inputs consistent. Next, images were then flattened into one-dimensional vectors to make it applicable to Random Forest, which needs a tabular structured input. Lastly, emotion category names were converted to integer labels from 0 to 6 to make it easier for sci-kit learn to distinguish between them.

2.3 Class Imbalance

An important characteristic of the FER-2013 dataset is its uneven class distribution. As shown in Figure 1, the happy category contains approximately 7,200 training images while disgust is only around 450, making it significantly smaller than the size of the happy emotion. This imbalance shapes the model's behavior which is why macro F1 was used instead of accuracy as the primary metric for model evaluation at the end. Using accuracy alone would introduce bias because of the underrepresented classes.

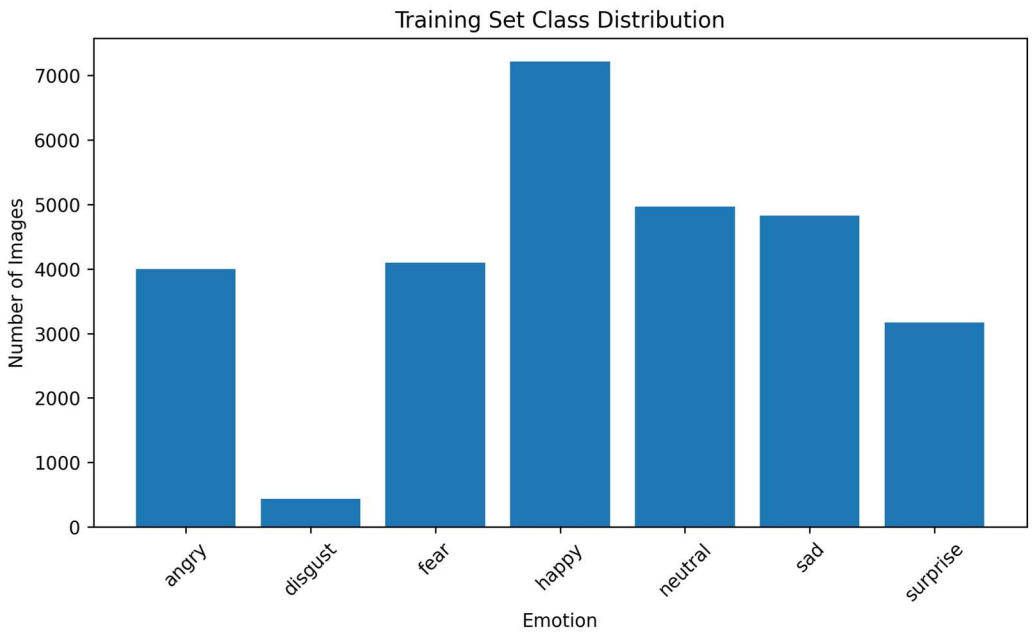

Figure 1. Training set class distribution across all seven emotion categories.

3. Modeling Methods

3.1 Random Forest Without PCA (Baseline)

The first method tested was a Random Forest classifier applied directly to the full flattened pixel feature set without any dimensionality reduction. Random Forest is an machine learning method that constructs many decision trees which are each trained on a random subset of sample/features. It then determines final predictions through majority vote of all the trees. We used 100 estimators with a maximum tree depth of 3 for the baseline. Limiting the tree depth to 3 levels (more than 2, less than 5) was a choice made to prevent individual trees from memorizing training data overtime. Since the feature space of the dataset is large relative to the number of distinct emotions, deeper trees would latch onto noise rather than patterns, which is not what we wanted. A more shallow depth of 3 forces each tree to focus only on the strongest splits at each node, and using 100 such trees combine these shallow patterns into a more stable overall prediction when trees are averaged together in the last step of Random Forest.

One constraint we faced with our implementation was the cost of running the full model on the complete training set. Running the full Random Forest on the complete training set of approximately 35,000 images with 2,304 features per image was very time intensive on the standard laptop hardware. Since training and prediction times were so long, experimenting more with hyperparameters settings was not feasible within the given project timeline. This limitation is discussed further in the closing section as an area for future improvement.

3.2 Random Forest With PCA

The second method combined PCA with the same Random Forest classifier. PCA is a linear transformation technique that identifies the directions of maximum variance in the data and transforms the original features into a smaller feature space only made of these principal components. By retaining only the most important components, PCA can reduce noise, decrease training time, and improve generalization of patterns which can be both good and bad.

The range of component values tested, (50, 100, 150, 200, 250, and 300), was determined by looking at the cumulative explained variance curve shown in Figure 2. The curve rises steeply and begins to flatten around 50 to

300 components, showing that this range captures the most explainable variance in our data. Testing within this specific window allowed us to evaluate whether the components captured by PCA were sufficient for classification without including the large number of supposedly less important components that follow. The PCA transformation was fitted on the training data and then applied to the test set to prevent data leakage and memorization of training data.

Both methods are reasonable for this classification task. Random Forest without PCA preserves the full details with the cost of not reducing dimensionality. However, Random Forest with PCA trades some of the specific detail for a cleaner and more compact representation of important features. When compared directly, we were able to answer the core research question: *Does compressing the feature space help or hurt emotion classification from raw pixel data?*

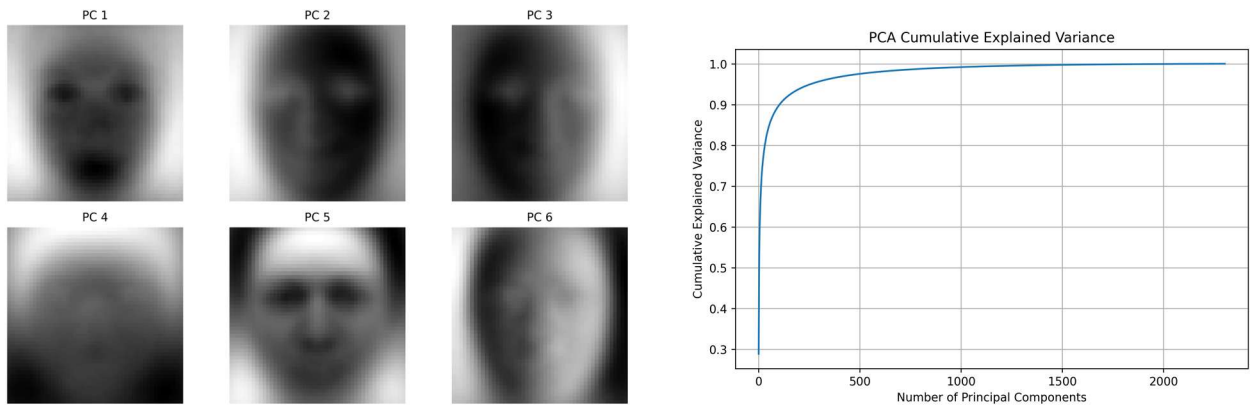

Figure 2. Top six PCA principal components visualized as eigenfaces (top) and cumulative explained variance curve (bottom).

Visualizing the first six principal components as eigenfaces provides insight into what information PCA captured from the dataset. As shown, PC 1 represents the overall facial brightness and the general shape of a face, which is clearly the most common pattern shared across all images. PC 2 and PC 3 capture more shading contrasts like differences in lighting between both sides of the face. By PC 4 through PC 6, the components begin looking at more details like lighting differences and slight orientation differences. One notable insight from our experiment is that it is difficult to see what exactly PCA is distinguishing by as the outputs of the principal components are in images, not specific values or text. We were able to conclude that none of these top principal components visibly isolate the features that distinguish emotions from one another in real life - such as the curvature of the lips, tension around the eyes, or raised eyebrows. This supports the conclusion that PCA was preserving general face structure rather than emotion-specific detail, which helps explain why classification performance was worse when these principal components were used as input to the Random Forest.

4. Results and Discussion

4.1 Evaluation Metrics

Model performance was evaluated using four metrics best suited for our dataset and problem. Accuracy measures the proportion of all test images classified correctly. However, because our dataset was imbalanced by the classes, we used Macro F1 which computes the F1 score for each emotion separately and then averages them equally. This ensures that every class/emotion was given the same weight despite number of images. We also used precision, which captures how often the model was correct predicting a given emotion. Finally, recall measures how many true examples of that emotion were successfully identified.

4.2 Accuracy and Macro F1 Comparison

The line charts in Figure 3 compare the PCA component counts against accuracy and macro F1 scores. The baseline Random Forest without PCA achieved an accuracy of 0.467 and a macro F1 score of 0.443, which is shown as the dashed horizontal line in both graphs. However, every PCA result fell below this baseline. For 50 components, the model with PCA reached its best result of 0.442 accuracy and 0.427 macro F1, which is already lower than the baseline. As component count increased by 100, 150, 200, 250, and then 300, both metrics declined, reaching 0.395 accuracy and 0.372 macro F1 at 300 components. This negative trend indicates that adding more principal components makes the performance of Random Forest classifier go down, since higher components means more bias/less variance and eventually less emotion indicative patterns.

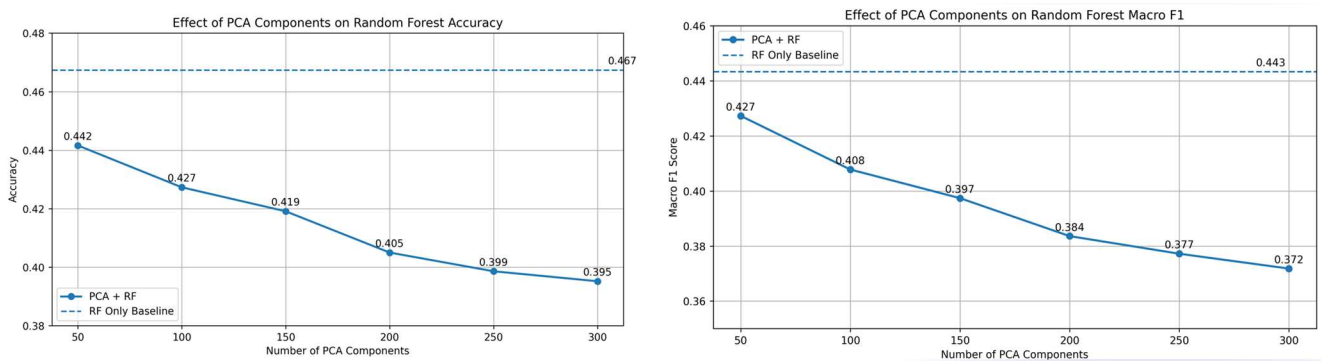

Figure 3. Effect of PCA component count on Random Forest accuracy (left) and macro F1 score (right).

4.3 Confusion Matrix Analysis

The confusion matrices in Figure 4 shown a closer view of where each model succeeds and fails. In the model with PCA, the happy column is saturated across nearly every row, meaning the model frequently predicted happy regardless of the true emotion. Neutral images were predicted as happy 615 times compared to only 371 correct neutral predictions. Sad images were predicted as happy 546 times versus 346 correct. This correlation matrix demonstrates how PCA compressed the feature space in a way that got rid of important distinctions between emotions that tend to look similar.

However, the baseline Random Forest without PCA shows a meaningfully better pattern. Neutral improved from 371 to 501 correct predictions, sad from 346 to 446, and fear from 250 to 296. Happy correct predictions decreased slightly from 1,447 to 1,379, suggesting the baseline had better predictions for categories that were harder to classify. Disgust was still weak in both models with only 35 and 32 correct predictions, but this makes sense because of its limited training data. Surprise performed well in both models, with 452 correct in PCA and 494 in the baseline, likely due to it being an emotion that is visually distinct open mouth and raised eyebrows.

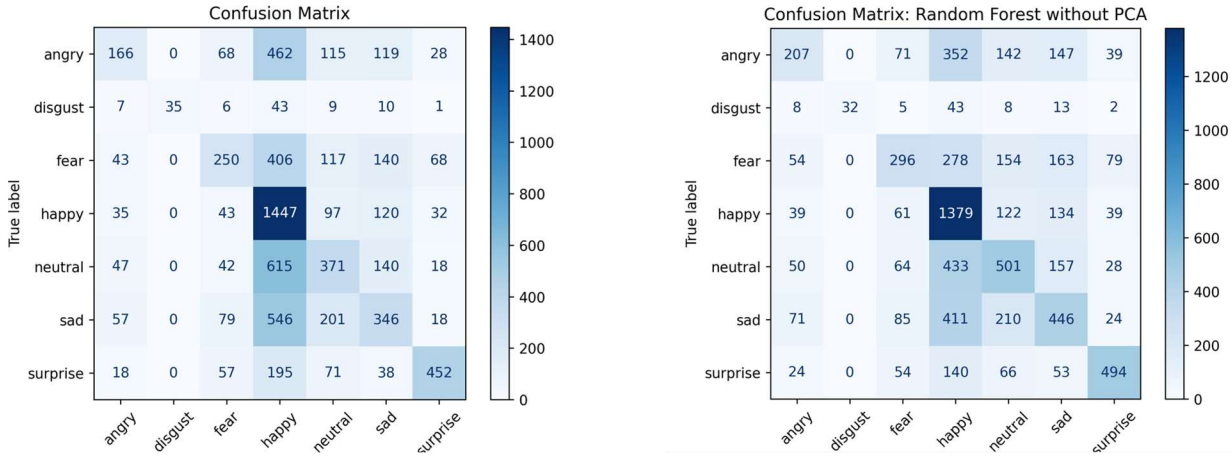

Figure 4. Confusion matrices for Random Forest with PCA (left) and without PCA (right).

4.4 Interpretation

The principal components captured global lighting and contrast differences, as visible in Figure 2, but ignored the smaller facial details that separate emotions from one another. The baseline model with just Random Forest retained all pixel information with the cost of higher dimensionality. Ultimately, this gave the Random Forest more patterns to work with, producing better outcomes across nearly every emotion category. The gap in the macro F1 score of 0.443 versus 0.372 at the worst PCA setting confirms that the baseline model with Random Forest outperformed with Random Forest model with PCA.

5. Closing Discussion

5.1 What Went Well

The steps used to create our project were organized and reproducible for different datasets. Because we are tested multiple PCA component values rather than a single configuration of parameters, we got a better picture of how dimensionality reduction affects performance across many compression levels. Using the cumulative explained variance curve to guide the selection of component values was an approach that helped us understand which values would be the most optimal in explaining differences in faces (variance between images). The use of macro F1 and accuracy provided a more complete view of model behavior given the class imbalance, and the confusion matrix analysis successfully identified which emotion pairs were most often confused and reasons as to why.

5.2 Areas for Improvement

However, there were several limitations of this project which offer clear directions for improvement. First, the class imbalance was identified but not corrected. Applying oversampling techniques such as SMOTE, which creates synthetic datapoints for underrepresented classes, could help improve performance on underrepresented emotions in our data like disgust. Since random Forest applies flattened pixel data and ignores spatial structure , a convolutional neural network is better for this task and usually outperforms traditional classifiers on image recognition tasks.

Computational constraints also were a significant challenge for this project. Training the Random Forest on roughly 35,000 images with 2,304 features per image was very slow on standard laptop hardware, to the point where prediction alone took a very long time. This forced the use of a shallow tree depth of 3 and limited the range of hyperparameter changes that could realistically be explored. Given more time and access to faster hardware, testing hyperparameters like deeper trees, more estimators, minimum samples per leaf, and features per split would likely yield better results.

Lastly, the 48x48 pixel resolution of the images is a constraint on what any model can detect. Increasing the image resolution would give both PCA and the Random Forest more pixels to find patterns in, which would help the model recognize facial features like lip curvature and brow tension that are difficult to distinguish at the current grid size. The tradeoff is that higher resolution increases feature count and training time, making this improvement more practical if access to advanced and higher power resources was available.

5.3 Future Research Directions

Overall, this project raises many interesting questions and directions for future research. The fundamental idea we explored is whether dimensionality and feature reduction methods are better than simpler modeling methods like Random Forest and the tradeoff between performance and computational cost. Examining whether training on a more balanced dataset using synthetic data generation leads to more consistent performance across all seven categories is another research direction that can be taken. Comparing this more traditional machine learning approach to a more advanced convolutional network architecture would help clarify the conditions under which

simpler methods remain competitive given real-world constraints on compute resources and training data availability. The core idea of this project can be tested with more real-world applications such as facial recognition algorithms, security and theft recognition systems, and more.